Calculus II Assignment 11

Part I. Regular Problems

1. Let D be the region in $\mathbb{R}^2$ where $0 \leq x$ and $0 \leq y \leq 1$.
   (a) Evaluate $\int_D \mathrm{e}^{-x}\,\mathrm{d}x\,\mathrm{d}y$
   (b) Evaluate $\int_D \mathrm{e}^{-x\sqrt{y}}\,\mathrm{d}x\,\mathrm{d}y$
   (c) Show that e^{-xy} is not integrable over D.
2. Let U be the open unit disk in $\mathbb{R}^2$ centered at the origin, $\sqrt{x^2+y^2} = r < 1$. Which of these functions are integrable over $\mathbb{R}^2 \setminus U := \{(x,y) \in \mathbb{R}^2 : (x,y) \notin U\}$?
   (a) $\log r$
   (b) r^{-2}
   (c) $r^{-2.1}$
   (d) r^{-3}
3. Evaluate the integrals:
   (a) $\int_0^1 \int_1^2 \int_2^3 \cos[\pi(x+y+z)]dxdydz$
   (b) $\int_0^1 \int_0^x \int_0^y (y+xz)dzdydx$
   (c) $\int_0^2 \int_0^x \int_0^{x+y} dzdydx$
4. Evaluate the triple integral $\iiint_E 6xydV$, where E lies under the plane $z = 1+x+y$ and above the region in the xy-plane bounded by the curves $y = \sqrt{x}, y = 0$, and $x = 1$.
5. Use a triple integral to find the volume of the given solid: the tetrahedron enclosed by the coordinate planes and the plane $2x + y + z = 4$.
6. Find the average value of the function $f(x,y,z) = xyz$ over the cube with side length L that lies in the first octant with one vertex at the origin and edges parallel to the coordinate axes.
7. Use cylindrical coordinates to evaluate the triple integral.
   (a) Evaluate $\iiint_E \sqrt{x^2+y^2}dV$, where E is the region that lies inside the cylinder $x^2 + y^2 = 16$ and between the planes $z = -5$ and $z = 4$.
   (b) Evaluate $\iiint_E x^2 dV$, where E is the solid that lies within the cylinder $x^2 + y^2 = 1$, above the plane $z = 0$, and below the cone $z^2 = 4x^2 + 4y^2$.

8. Sketch the solid whose volume is given by the integral and evaluate the integral:

$$\int_0^{\pi/6} \int_0^{\pi/2} \int_0^3 \rho^2 \sin\phi d\rho d\theta d\phi.$$

9. Find the volume of the solid that lies within the sphere $x^2 + y^2 + z^2 = 4$, above the xy-plane, and below the cone $z = \sqrt{x^2 + y^2}$.

10. Find the volume of the solid E that lies above the cone $z = \sqrt{x^2 + y^2}$ and below the sphere $x^2 + y^2 + z^2 = 1$. [**Hint:** use cylindrical or spherical coordinates, whichever seems more appropriate.]

Part II. Group Problems

1. Perform the indicated integration over the given box:
   (a) $\iiint_B x^2 dxdydz, B = [0,1] \times [0,1] \times [0,1]$
   (b) $\iiint_B e^{-xy} y dxdydz, B = [0,1] \times [0,1] \times [0,1]$
   (c) $\iiint_B (2x + 3y + z) dxdydz, B = [0,2] \times [-1,1] \times [0,1]$